

CENTRAL BOARD OF SECONDARY EDUCATION

SAMPLE QUESTION PAPER

Academic Session: 2026–27

Class: 10

Subject: Mathematics

Maximum Marks: 60

Time Allowed: 2 Hours

GENERAL INSTRUCTIONS

- This Question Paper has 5 Sections A, B, C, D, and E.
- Section A has 10 Multiple Choice Questions (MCQs) of 1 mark each.
- Section B has 4 Very Short Answer (VSA) Questions of 2 marks each.
- Section C has 4 Short Answer (SA) Questions of 3 marks each.
- Section D has 2 Long Answer (LA) Questions of 5 marks each.
- Section E has 2 Case Study Based Questions of 4 marks each.
- All questions are compulsory. However, an internal choice in 2 Qs of 2 marks, 2 Qs of 3 marks and 2 Qs of 5 marks has been provided.
- Draw neat figures wherever required. Take $\pi = 22/7$ wherever required unless stated.

SECTION A

Multiple Choice Questions

Q1. If the sum of the zeroes of the quadratic polynomial $P(x) = kx^2 + 2x + 3k$ is equal to their product, then the value of k is:

(A) $-2/3$

(B) $2/3$

(C) $1/3$

(D) $-1/3$

(1 Mark)

Q2. The pair of equations $y = 0$ and $y = -7$ has:

(A) One solution

(B) Two solutions

(C) Infinitely many solutions

(D) No solution

(1 Mark)

Q3. The 17th term of an AP exceeds its 10th term by 7. The common difference is:

(A) 1

(B) 2

(C) 3

(D) 4

(1 Mark)

Q4. For what value(s) of p does the quadratic equation $4x^2 - 12x + p = 0$ have real and equal roots?

(A) $p < 9$

(B) $p > 9$

(C) $p = 9$

(D) $p = 0$

(1 Mark)

Q5. In $\triangle ABC$, D and E are points on sides AB and AC respectively such that $DE \parallel BC$. If $AD/DB = 3/4$ and $AC = 14$ cm, then AE is:

(A) 6 cm

(B) 7 cm

(C) 8 cm

(D) 9 cm

(1 Mark)

Q6. If $\tan \theta = 4/3$, then the value of $(\sin \theta + \cos \theta) / (\sin \theta - \cos \theta)$ is:

(A) 7

(B) $1/7$

(C) $4/3$

(D) $3/4$

(1 Mark)

Q7. The distance between the points $(a \cos \theta + b \sin \theta, 0)$ and $(0, a \sin \theta - b \cos \theta)$ is:

(A) $a^2 + b^2$

(B) $\sqrt{a^2 + b^2}$

(C) $a^2 - b^2$

(D) $\sqrt{a^2 - b^2}$

(1 Mark)

Q8. A card is drawn from a well-shuffled deck of 52 cards. The probability that the card drawn is not an ace is:

(A) $1/13$

(B) $4/13$

(C) $12/13$

(D) 1

(1 Mark)

Q9. Assertion (A): The HCF of two numbers is 18 and their product is 3072. Then their LCM is 170.66.

Reason (R): For any two positive integers a and b, $\text{HCF}(a, b) \times \text{LCM}(a, b) = a \times b$.

(A) Both A and R are true and R is the correct explanation of A.

(B) Both A and R are true but R is not the correct explanation of A.

(C) A is true but R is false.

(D) A is false but R is true.

(1 Mark)

Q10. Assertion (A): The length of the tangent from an external point P to a circle with centre O and radius r is always greater than r .

Reason (R): The tangent at any point of a circle is perpendicular to the radius through the point of contact.

(A) Both A and R are true and R is the correct explanation of A.

(B) Both A and R are true but R is not the correct explanation of A.

(C) A is true but R is false.

(D) A is false but R is true.

(1 Mark)

SECTION B

Very Short Answer Questions

Q11. Find the largest number that divides 245 and 1029 leaving a remainder of 5 in each case.

(2 Marks)

Q12. If the points A(6, 1), B(8, 2), C(9, 4) and D(p, 3) are the vertices of a parallelogram, taken in order, find the value of p.

OR

Find the ratio in which the line segment joining the points (-3, 10) and (6, -8) is divided by (-1, 6).

(2 Marks)

Q13. In the given figure, if TP and TQ are the two tangents to a circle with centre O so that $\angle POQ = 110^\circ$, then find $\angle PTQ$.

(Diagram: A circle with center O. Points P and Q are on the circle. TP and TQ are tangents from an external point T. OP and OQ are radii. Angle POQ is 110 degrees.)

(2 Marks)

Q14. A die is thrown once. Find the probability of getting:

(i) a prime number

(ii) a number lying between 2 and 6

OR

A bag contains 5 red balls and some blue balls. If the probability of drawing a blue ball is double that of a red ball, find the number of blue balls in the bag.

(2 Marks)

SECTION C

Short Answer Questions

Q15. Prove that $\sqrt{5}$ is an irrational number.

(3 Marks)

Q16. Find the zeroes of the quadratic polynomial $6x^2 - 3 - 7x$ and verify the relationship between the zeroes and the coefficients.

(3 Marks)

Q17. The sum of the first n terms of an AP is given by $S_n = 2n^2 + 3n$. Find the 10th term of the AP.

OR

If the 9th term of an AP is zero, prove that its 29th term is twice its 19th term.

(3 Marks)

Q18. Prove that: $(\sin A + \operatorname{cosec} A)^2 + (\cos A + \sec A)^2 = 7 + \tan^2 A + \cot^2 A$.

OR

If $\sin \theta + \cos \theta = \sqrt{3}$, then prove that $\tan \theta + \cot \theta = 1$.

(3 Marks)

SECTION D

Long Answer Questions

Q19. A train travels 360 km at a uniform speed. If the speed had been 5 km/h more, it would have taken 1 hour less for the same journey. Find the speed of the train.

OR

A motor boat whose speed in still water is 18 km/h, takes 1 hour more to go 24 km upstream than to return downstream to the same spot. Find the speed of the stream.

(5 Marks)

Q20. A solid is in the shape of a cone standing on a hemisphere with both their radii being equal to 1 cm and the height of the cone is equal to its radius. Find the volume of the solid in terms of π .

OR

A metallic sphere of radius 4.2 cm is melted and recast into the shape of a cylinder of radius 6 cm. Find the height of the cylinder.

(5 Marks)

SECTION E

Case Study

Q21. Case Study 1: Lighthouse Observation

A lighthouse is a tall tower with a light at the top, used as an aid to navigation for maritime pilots at sea or on inland waterways. The light from a lighthouse can be seen from a certain distance.

A person is standing on the deck of a ship, which is 10 m above water level. He observes that the angle of elevation of the top of a lighthouse is 60° and the angle of depression of the base of the lighthouse is 30° .

Based on the above information, answer the following questions:

- (i) Draw a neat labelled diagram to represent the given situation.
- (ii) Find the height of the lighthouse.
- (iii) Find the distance of the ship from the base of the lighthouse.

(4 Marks)

Q22. Case Study 2: Sports Day Event

To conduct Sports Day activities in your school, lines have been drawn with chalk powder at a distance of 1 m each, in a rectangular field ABCD, 100 flower pots have been placed at a distance of 1 m from each other along AD, as shown in the figure. Niharika runs $\frac{1}{4}$ th the distance AD on the 2nd line and posts a green flag. Preet runs $\frac{1}{5}$ th the distance AD on the 8th line and posts a red flag.

(Diagram: A rectangular field ABCD. Lines are drawn parallel to AB at 1m intervals. Flower pots are placed along AD at 1m intervals. Niharika is on the 2nd line, Preet on the 8th line.)

Based on the above information, answer the following questions:

- (i) What are the coordinates of the green flag and the red flag?
- (ii) Find the distance between the two flags.
- (iii) If Rashmi has to post a blue flag exactly halfway between the line segment joining the two flags, where should she post her flag?

(4 Marks)

End of Question Paper